Лабораторная №5
Управление системными службами

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Цель работы

Получить навыки управления системными службами операционной системы посредством systemd.

Рабочая машина

Процессор: Ryzen 7 7435hs 3.10 GHz
Оперативка: DDR5 16x2 4800 MHz
Видеокарта: RTX 4050 6 GB GDDR6

Ход выполнения лабораторной работы

```
adskalozub@localhost:~$ sudo -i
[sudo] пароль для adskalozub:
root@localhost:~# systemctl status vsftpd
Unit vsftpd.service could not be found.
root@localhost:~# dnf -y install vsftpd
```

Рис 1. устанавливаем vsftpd

```
root@localhost:~# systemctl start vsftpd
root@localhost:~# systemctl status vsftpd
● vsftpd.service - Vsftpd ftp daemon
     Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; disabled; preset: dis>
     Active: active (running) since Mon 2026-03-09 19:29:02 MSK; 14s ago
 Invocation: 91fd17b3162b4b44821aee2da81ef12f
    Process: 8460 ExecStart=/usr/sbin/vsftpd /etc/vsftpd/vsftpd.conf (code=exited,>
   Main PID: 8461 (vsftpd)
      Tasks: 1 (limit: 48470)
     Memory: 856K (peak: 1.3M)
        CPU: 8ms
     CGroup: /system.slice/vsftpd.service
             └─8461 /usr/sbin/vsftpd /etc/vsftpd/vsftpd.conf

мар 09 19:29:02 localhost.localdomain systemd[1]: Starting vsftpd.service - Vsftpd>
мар 09 19:29:02 localhost.localdomain systemd[1]: Started vsftpd.service - Vsftpd >
lines 1-14/14 (END)
```

Рис 2. смотрим статус vsftpd

```
root@localhost:~# ls /etc/systemd/system/multi-user.target.wants
atd.service           firewalld.service       remote-cryptsetup.target
auditd.service        irqbalance.service      remote-fs.target
audit-rules.service   kdump.service           rsyslog.service
avahi-daemon.service  libstoragemgmt.service  smartd.service
chronyd.service       mcelog.service          sshd.service
crond.service         mdmonitor.service       sssd.service
cups.path             ModemManager.service    tuned.service
cups.service          NetworkManager.service  vmtoolsd.service
root@localhost:~# systemctl enable vsftpd
Created symlink '/etc/systemd/system/multi-user.target.wants/vsftpd.service' → '/us
r/lib/systemd/system/vsftpd.service'.
root@localhost:~# systemctl status vsftpd
● vsftpd.service - Vsftpd ftp daemon
     Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: disa>
     Active: active (running) since Mon 2026-03-09 19:29:02 MSK; 1min 46s ago
 Invocation: 91fd17b3162b4b44821aee2da81ef12f
   Main PID: 8461 (vsftpd)
      Tasks: 1 (limit: 48470)
     Memory: 856K (peak: 1.3M)
        CPU: 8ms
     CGroup: /system.slice/vsftpd.service
             └─8461 /usr/sbin/vsftpd /etc/vsftpd/vsftpd.conf

мар 09 19:29:02 localhost.localdomain systemd[1]: Starting vsftpd.service - Vsftpd>
мар 09 19:29:02 localhost.localdomain systemd[1]: Started vsftpd.service - Vsftpd >
lines 1-13/13 (END)
```

Рис 3. подключение systemctl

```
root@localhost:~# systemctl list-dependencies vsftpd
vsftpd.service
```

Рис 4. выводим systemctl

```
root@localhost:~# systemctl list-dependencies vsftpd --reverse
vsftpd.service
● └multi-user.target
●   └graphical.target
```

Рис 5. выводим systemctl

```
root@localhost:~# dnf -y install iptables\*
```

Рис 6. устанавливаем необходимые пакеты

```
root@localhost:~# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
     Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: e>
     Active: active (running) since Mon 2026-03-09 17:06:02 MSK; 2h 26min ago
 Invocation: abe1340c57f1402ba5ee31778c4c5ec6
       Docs: man:firewalld(1)
   Main PID: 1151 (firewalld)
      Tasks: 2 (limit: 48470)
     Memory: 49.6M (peak: 72.3M)
        CPU: 917ms
     CGroup: /system.slice/firewalld.service
             └1151 /usr/bin/python3 -sP /usr/sbin/firewalld --nofork --nopid

мар 09 17:06:00 localhost systemd[1]: Starting firewalld.service - firewalld - dyn>
мар 09 17:06:02 localhost systemd[1]: Started firewalld.service - firewalld - dyna>

root@localhost:~# systemctl status iptables
○ iptables.service - IPv4 firewall with iptables
     Loaded: loaded (/usr/lib/systemd/system/iptables.service; disabled; preset: d>
     Active: inactive (dead)
```

Рис 7. выводим статус

```
root@localhost:~# systemctl start firewalld
root@localhost:~# systemctl start iptables
```

Рис 8. запускаем службы

```
root@localhost:~# cat /usr/lib/systemd/system/firewalld.service
[Unit]
Description=firewalld - dynamic firewall daemon
```

Рис 9. смотрим параметры

```
root@localhost:~# cat /usr/lib/systemd/system/iptables.service
[Unit]
Description=IPv4 firewall with iptables
```

Рис 10. смотрим параметры

```
root@localhost:~# systemctl stop iptables
root@localhost:~# systemctl start firewalld
root@localhost:~# systemctl mask iptables
Created symlink '/etc/systemd/system/iptables.service' → '/dev/null'.
root@localhost:~# systemctl start iptables
Failed to start iptables.service: Unit iptables.service is masked.
root@localhost:~# systemctl enable iptables
Failed to enable unit: Unit /etc/systemd/system/iptables.service is masked
```

Рис 11. редактируем параметры systemctl

```
root@localhost:~# systemctl --type=target
  UNIT                      LOAD   ACTIVE SUB    DESCRIPTION
  basic.target              loaded active active Basic System
```

Рис 12. смотрим параметры

```
root@localhost:~# systemctl --type=target --all
  UNIT
  basic.target
```

Рис 13. смотрим параметры

```
adskalozub@localhost:~$ sudo -i
[sudo] пароль для adskalozub:
root@localhost:~# systemctl get-default
graphical.target
root@localhost:~# systemctl set-default
Too few arguments.
root@localhost:~# systemctl set-default multi-user.target
Removed '/etc/systemd/system/default.target'.
Created symlink '/etc/systemd/system/default.target' → '/usr/lib/systemd/system/mul
ti-user.target'.
root@localhost:~# systemctl set-default graphical.target
Removed '/etc/systemd/system/default.target'.
Created symlink '/etc/systemd/system/default.target' → '/usr/lib/systemd/system/gra
phical.target'.
root@localhost:~#
```

Рис 14. выстав параметры

Вывод

Мы научились работать со службами